INTRODUCTION

Option contracts are contingent claims in which one of the counterparties determines whether and when a trade will settle. Unlike a forward commitment with a value of zero to both counterparties at inception, an option buyer pays a premium to the seller for the right to transact the underlying in the future at a pre-agreed price. The contingent nature of options affects their price as well as their value over time.

In the first lesson, we explore three features unique to contingent claims related to an option's value versus the spot price of the underlying: the exercise, or intrinsic, value; the relationship between an option's spot price and its exercise price, referred to as "moneyness"; and the time value. We then turn to how the arbitrage and replication concepts introduced earlier for forward commitments differ when applied to contingent claims with an asymmetric payoff profile. Finally, we identify and describe factors that determine the value of an option. These lessons focus on European options, which can be exercised only at expiration.

Category

Pricing and Valuation of Options

Learning Module Overview

- An option's value comprises its exercise value and its time value. The exercise value is the option's value if it were *immediately* exercisable, while the time value captures the *possibility* that the passage of time and the variability of the underlying price will increase the profitability of exercise at maturity.
- Option moneyness expresses the relationship between the underlying price and the exercise price. A put or call option is "at the money" when the underlying price equals the exercise price. An option is more likely to be exercised if it is "in the money"—with an underlying price above (for a call) or below (for a put) the exercise price—and less likely to be exercised if it is "out of the money."
- Due to their asymmetric payoff profile, options are characterized by no-arbitrage price bounds. The lower bound is a function of the present value of the exercise price and the underlying price, while the upper bound is the underlying price for a call and the exercise price for a put.
- As in the case of forward commitments, the replication of option contracts uses a combination of long (for a call) or short (for a put) positions in an underlying asset and borrowing or lending cash. The replicating transaction for an option is based on a proportion of the underlying, which is closely associated with the moneyness of the option.
- The underlying price, the exercise price, the time to maturity, the risk-free rate, the volatility of the underlying price, and any income or cost associated with owning the underlying asset are key factors in determining the value of an option.
- Changes in the volatility of the underlying price and the time to expiration will usually have the same directional effect on put and call option values. Changes to the exercise price, the risk-free rate, and any income or cost associated with owning the underlying asset have the opposite effect on call options versus put options.

Learning Module Self-Assessment

understanding of this learning module.

1. Which of the following statements correctly describes the lower bound of a call option's value?

A. The underlying's price minus the present value of the option's exercise price
B. The underlying's price minus the option's exercise price or zero, whichever is greater
C. The underlying's price minus the present value of the option's exercise price or zero, whichever is greater

Hide Solution

Solution:

C is correct. The lower bound of a call price is the underlying's price minus the present value of its exercise price or zero, whichever is greater. A is incorrect as the response omits the fact that the lower bound is zero if the underlying's price is less than the present value of the exercise price. B is incorrect as it omits the present value term.

2. Match the following statements about replication strategies with their associated derivative instrument(s):

1. At time $t = 0$, lend at the risk-free rate and sell the underlying at S_0.	A. Neither a call option nor a put option replication strategy
2. The replication strategy is executed at inception and is settled at maturity with no adjustment over time.	B. A put option replication strategy
3. At time $t = T$, sell the underlying at S_T and repay the loan of X.	C. A call option replication strategy if exercised

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Solution:

1. B is correct. At time $t = 0$, a put option replication strategy involves lending at the risk-free rate and selling the underlying at S_0.
2. A is correct. As both call and put options involve a non-linear payoff profile, their replication strategy requires adjustment over time as the likelihood of exercise changes.
3. C is correct. A call option replication strategy if exercised involves repaying the loan of X and selling the underlying at S_T at time $t = T$.

3. A European call option with three months remaining to maturity on an underlying stock with no additional cash flows has an exercise price (X) of GBP 50, a risk-free rate of 2%, and a current underlying price (S_t) of GBP 57.50. If the current call option price is GBP 10, which response below most closely shows the correct exercise value and the time value of the option?

A. Exercise value = GBP 7.50; Time value = GBP 2.50
B. Exercise value = GBP 7.75; Time value = GBP 2.25
C. Exercise value = GBP 0; Time value = GBP 10

Hide Solution

Solution:

B is correct. An option's value comprises its exercise value plus its time value. The exercise value of a call option is Max (0, S_t − PV(X)) and is calculated as follows:

$$\text{Call Option Exercise Value} = \text{Max}\left(0,\ S_t - X(1+r)^{-(T-t)}\right)$$

$$\text{Max}(0,\ \text{GBP } 57.50 - \text{GBP } 50(1.02)^{-0.25})$$

present value of the exercise price. The time value is the difference between the option price and the exercise value, representing the possibility that the option payoff at maturity will exceed the current exercise value due to a favorable price change:

$$\text{Call Option Time Value} = c_t - Max\left(0,\ S_t - X(1+r)^{-(T-t)}\right)$$

= GBP 2.25 (= GBP 10 − GBP 7.75)

The time value is always positive and declines to zero at maturity ($t = T$). A is incorrect as the present value term is omitted from the exercise value calculation. C is incorrect as it implies the option has zero exercise value.

4. Match the following underlying price and exercise price relationships with their associated put option:

1. $S_T = 100$, $X = 100$	A. An at-the-money put option
2. $S_T = 110$, $X = 100$	B. An in-the-money put option
3. $S_T = 90$, $X = 100$	C. An out-of-the-money put option

Hide Solution

Solution:

Put options are in the money when $S_T < X$, at the money when $S_T = X$, and out of the money when $S_T > X$. Therefore:

1. A is correct. Since $S_T = X = 100$, this is an at-the-money put option.
2. C is correct. Since $S_T > X$, this is an out-of-the-money put option.
3. B is correct. Since $S_T < X$, this is an in-the-money put option.

5. Match the following changes in a factor affecting option value (holding other factors constant) with their corresponding option value change:

1. A higher exercise price (X)	A. Decreases the value of both a call option and a put option
2. A higher underlying price (S_T)	B. Decreases the value of a call option
3. A decline in the volatility of the underlying price	C. Decreases the value of a put option

Hide Solution

Solution:

1. B is correct. A higher exercise price decreases the value of a call option; for a given underlying price at maturity (S_T), the call option settlement value of Max (0, $S_T - X$) will decrease for a higher X.
2. C is correct. A higher underlying price (S_T) will decrease the value of a put option. Since a put option is the right to sell an underlying, the put option settlement value of Max (0, $X - S_T$) will fall as S_T rises.
3. A is correct. A decline in the volatility of the underlying price will decrease the value of both a call option and a put option. Lower price variability of the underlying will reduce the probability of a higher positive exercise value for a call or a put option without affecting the downside case where the option expires unexercised.

6. Which of the following statements provides the correct description as to how a call option's value changes if the income on the underlying declines unexpectedly, holding all else equal.

A. The call option value increases.
B. The call option value decreases.

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Solution:

A is correct. Income or other, non-cash benefits (such as convenience yield) accrue to the owner of an underlying asset but not to the owner of a derivative, whose value is based on the underlying. A call option on an underlying with income has lower value than an identical call option on the same underlying without income. Thus, a decline in income on an underlying asset increases the value of a call option.

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